[METRIC]
A-A-1770A
August 3, 1993
SUPERSEDES
A-A-1770
February 12, 1981

COMMERCIAL ITEM DESCRIPTION

FACESHIELD, INDUSTRIAL

This Commercial Item Description has been approved for use by the General Services Administration, Federal Supply Service.

1. SCOPE

1.1 Scope. This Commercial Item Description (CID) describes an industrial "flip-up" type faceshield with a crown protector and an adjustable headgear.

2. APPLICABLE DOCUMENTS

2.1 The documents referenced in this Commercial Item Description shall be the issues in effect on the date of the invitation for bids or request for proposal unless otherwise specified. These documents form a part of this Commercial Item Description to the extent specified. In the event that there is a conflict between this Commercial Item Description and a document referenced herein, this Commercial Item Description shall take precedence.

Federal Standard:

FED-STD-376 - Preferred Metric Units For General Use By The Federal Government.

Copies of Federal Standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any information which might improve this document should be sent to: General Services Administration, Engineering Group (7FXEE), 819 Taylor Street, Fort Worth, TX 76102.

DISTRIBUTION STATEMENT A: Approved for public release; distribution unlimited.

FSC 4240

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Copies of Military standards may be obtained from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

ASTM standard:

E 29 - Standard Practice For Using Digits In Test Data To Determine Conformance With Specifications.

Application for copies of ASTM standards should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.

ANSI standard:

Z87.1 - American National Standard Practice for Occupational and Educational Eye and Face Protection.

Application for copies of ANSI standards should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.

3. REQUIREMENTS

3.1 Design. As a minimum, faceshields furnished under this commercial item description shall conform to the requirements of ANSI Z87.1 for faceshields with non-removable windows. The faceshield shall have a headgear system for wearing on the head. There shall be a crown protector above the clear window. The entire window and crown protector/window unit shall be capable of rotating up out of the wearer's visual way by pivot points on the headgear placed approximately over each ear when properly worn. The crown protector/window shall be limited in travel and retained in the up and down positions at the pivot points. Whether or not the window is actually removable is at the contractor's option.

3.2 Material. The faceshield shall be composed of non-sparking and non-conductive materials. The portions that will be in contact with the wearer shall be composed of materials non-irritating to the skin.

3.3 Window. The window shall be clear and colorless. The size of the window free of the frame shall be 39.0 cm minimum side to side, measured at the point of attachment to the crown protector, and 19.0 cm minimum high overall, measured in the center of the window. The window thickness shall be sufficient to meet the applicable requirements of ANSI Z87.1, but shall be no less than 1.25 mm.

3.4 Crown protector. The crown protector shall be the portion that actually is attached to the pivot points and in turn supports the window.

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3.5 Headgear. The headgear shall be flexible and comfortable to the wearer. By comfortable, it is intended that the headgear shall present no sharp edges, corners or points to the wearer's head. The circumference size shall be adjustable in a range that includes as a minimum standard size 6-1/2 to size 8. This shall be adjustable using one hand with the headgear in place on the head. The headgear shall have at least one adjustable part or strap that crosses over the top of the head from side to side. The headgear shall be adjustable to firmly hold the faceshield assembly on the wearer's head without slipping or changing sizes during use once properly adjusted. The headgear shall be capable of flexing without damage when tested as specified in paragraph 4.3.2. The front of the headgear shall have an absorbent sweatband of at least 22 cm in length. The sweatband shall completely cover the portion of the inside headband to which it is attached when the headgear is worn.

3.6 Pivots. The pivot points shall retain the window at the top of its travel out of the wearer's way; and at the bottom of its travel, in the use position. The method of retention shall be integral with both pivots and shall not require any action on the wearer's part for normal engagement or release other than movement of the window. Hold points at other than the full up and full down positions are at the option of the contractor. The pivots shall be capable of withstanding 400 complete cycles without damage and still hold the crown protector/window in the up position when tested as specified in 4.3.3. If the pivot points can be adjusted without tools while on the wearer's head, tightening of the adjustment is permitted as necessary at any time to meet the requirements.

3.7 Workmanship. The faceshield shall be free from any cracks, fractures, or defects which may affect serviceability, durability or appearance.

3.8 Regulatory requirements. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

3.9 Special marking. Each faceshield and window shall be marked as required by ANSI Z87.1 (see 5.2).

3.10 Inch/pound products. Products manufactured to inch/pound dimensions will be considered on an equal basis with those manufactured using metric units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

3.11 Commercial item. The use of the term "commercial item description" in this document does not imply that any item or items offered are not required to conform with all requirements specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this Commercial Item Description where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Visual and dimensional examination. Sampling shall be in accordance with MIL-STD-105. The lot shall be all faceshields offered for inspection at one time. The sample unit shall be one complete, fully assembled faceshield as defined by paragraph 3.1. Sampling for visual and dimensional inspection shall be inspection level I, with an Acceptable Quality Level (AQL) of 2.5 percent defective. Each of the sample units shall be visually and dimensionally examined for conformance to the requirements of this Commercial Item Description. Tolerance limits specified are absolute limits as defined in ASTM E 29.

4.3 Tests. Sampling for testing shall be 4 complete faceshields from each lot. Failure of any sample shall reject the lot.

4.3.1 Industrial standards conformance. Unless otherwise specified, certificates of conformance stating that the offered items meet the requirements of ANSI Z87.1 are acceptable. Certification by the Safety Equipment Institute (SEI) as meeting the requirements of ANSI Z87.1 is acceptable as proof of conformance to the ANSI Z87.1 requirements.

4.3.2 Flex test. Two samples from each lot shall be used for this test. The sample headgear shall be twisted so the portion of the headband at the pivot points are 90 degrees to each other, at a room temperature of between 65 degrees Fahrenheit and 75 degrees Fahrenheit, with no breaks or fractures occurring. To perform the test, remove the window/crown protector portion. Each hand shall grasp the headgear at a pivot point. The hands shall be rotated in opposite directions so that the headbands at each pivot point are at 90 degrees from each other, held for 10 seconds and released. The sample shall be inspected and failed if any cracks, tears, or evidence of permanent damage, deformation, fracture, or weakness is visible. Repairable separation of the headgear components is acceptable. If the window/crown protector is not removable, the test shall be performed with it in place and any damage to it shall be ignored.

4.3.3 Pivot hold and wear tests. If the samples tested in 4.3.2 are undamaged, they may be used for the following tests, otherwise, the remaining two selected samples shall be tested. The following is a two-part test performed on the same samples. Failure of the hold test at any time before, during, or after the wear test fails the sample. If the pivots have a tensioning device that can be adjusted without the use of tools, the tension can be tightened at any time as necessary to pass the tests, however, at no time shall the tension be loosened.

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4.3.3.1 Hold test. On completely assembled faceshield test samples, firmly and rigidly attach a 100 gm (+/- 0.5 gm) weight to the lower most front of the window. The down position shall be defined as position of use, with the window down and against the stop. The up position shall be defined as the position where the window is rotated up at least 90° from the down position. With the window in the fully up position, hold only the headgear level with both hands without supporting the crown protector/window support. The window shall remain in the full up position. With the weight still attached and with the hands touching only the headgear portion, rotate the entire headgear/faceshield assembly slowly 360 degrees, once clockwise and once counterclockwise, about an axis extending approximately through the pivot points. The window and headgear must rotate as one unit without the window slipping at the pivot. Any rotation beyond the stop(s) of the pivot, breakage of any part of the faceshield, the appearance of cracks, or slippage of the locking mechanism shall fail the sample.

4.3.3.2 Wear test. Remove the weight from the window of the test samples. Supporting the headgear, the window shall be moved a total of 400 cycles at a rate of 21 (+/-4) cycles per minute. A single cycle shall be defined as the 90-degree (minimum) rotation from the full down position, to the full up position, and back to the original full down position. Once the 400 cycles have been completed, repeat the entire Pivot Hold Test. Samples that cannot pass the pivot hold test after 400 cycles shall fail. Any damage to the faceshield assembly caused by the wear test shall fail the sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The packaging, packing, and marking shall be as specified in the contract or order.

5.2 Special marking. The faceshield shall be marked with the manufacturer's name or recognized trademark. If the window is removable, both the window and the headgear/crown protector shall be so marked. The faceshield shall bear evidence that it meets the applicable requirements of ANSI Z87.1 and be marked as required by ANSI Z87.1.

6. NOTES

6.1 Intended use. The faceshields covered by this Commercial Item Description are intended for secondary facial protection against small flying particles and liquid splash. They are not intended to provide glare protection or as primary protection and should always be used in conjunction with approved primary eye protection.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number and date of this commercial item description.
b. The packaging, packing, and marking desired.
c. If lot by lot testing for industrial standard conformance is required.

6.3 NSN information. The only known NSN referencing this document is 4240-00-542-2048.

MILITARY INTERESTS:

Military Coordinating Activity:
Army - EA

Custodians:
Army - EA
Air Force - 99

Review Activities:
Army - ME
Air Force - 99

User Activities:
Army - MD
Navy - YD, MC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
GSA-FSS

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